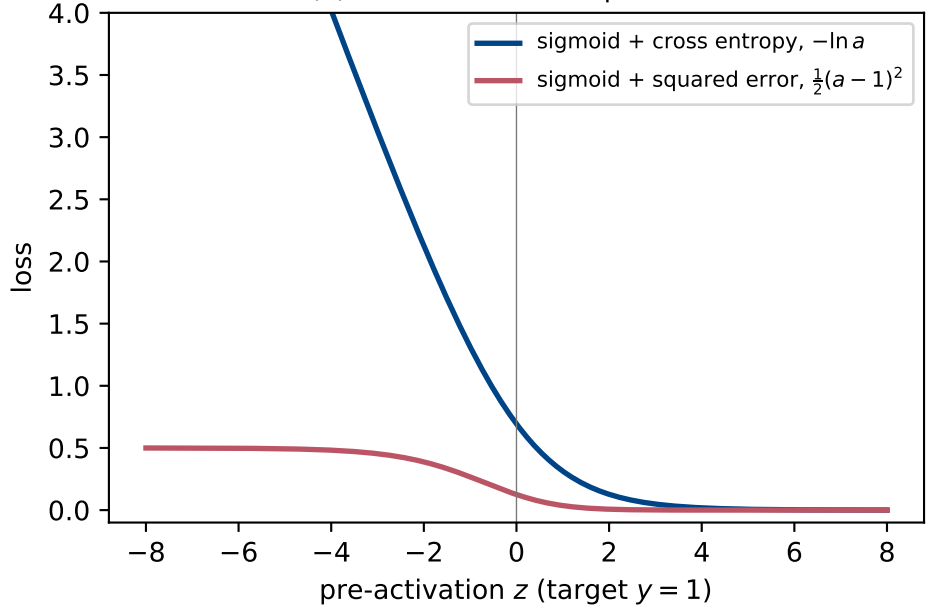
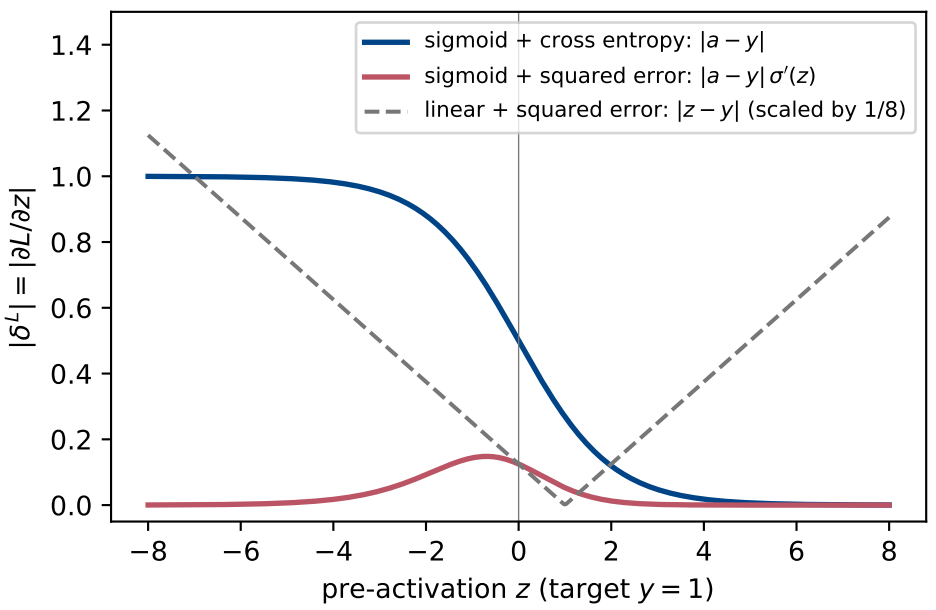

(a) the loss of one output node
loss
pre-activation z (target $y = 1$)
sigmoid + cross entropy, $-\ln a$
sigmoid + squared error, $\frac{1}{2}(a-1)^2$


(b) the output error δ^L that starts the backward pass
$|\delta^L| = |\partial L/\partial z|$
pre-activation z (target $y = 1$)
sigmoid + cross entropy: $|a-y|$
sigmoid + squared error: $|a-y|\,\sigma'(z)$
linear + squared error: $|z-y|$ (scaled by 1/8)